K. Change Test Record

Appendix K of the EFC Gases & Advanced Materials Controls Standard, Revision H, October 2, 2026. Standalone copy; see the full document for the other chapters and appendices.

This form documents and tests a change to a system that has already been built (Section 7.6 of the Controls Standard). Use one form per change. Scale the test steps to the change: a screen text edit may need one line, while an interlock change needs a full functional test. Changes to interlocks and safety-related items require an approved MOC (Section 7.3 of the Controls Standard) and, under process safety management, a pre-startup safety review before the equipment returns to service.[1]

K.1 Change Identification

Change / MOC no.		**Date**	
Project no. and system		**Requested by**	
Software before (file name)			
Software after (file name)			

File names follow Section 7.2 of the Controls Standard (for example `014_R23-Column-PLC_1`).

K.2 Change Type

☐ I/O (added, removed, or reassigned)	☐ Electrical (wiring, devices, power)	☐ Sequence
☐ Interlock or permissive	☐ Setpoint limit (permissive/protective)	☐ Alarm
☐ SCADA screen or tag	☐ Network	☐ Other: _____

MOC required? ☐ Yes (interlock, protective setpoint, permissive limit, bypass, E-stop or safety circuit) ☐ No

[1] OSHA 29 CFR 1910.119(l) (management of change) and (i) (pre-startup safety review); IEC 61511-1:2016+A1:2017 (modification of safety instrumented functions); ISA/IEC 62443-2-1:2024, SPE 2 (change control); IEC 60204-1:2016+A1:2021, 18.7 and NFPA 79-2024, 18.7 (retesting after modification).

K.3 Description and Impact

What is changing and why	
Affected equipment	
Affected interlocks, permissives, and alarms	
Affected drawings and documents	

K.4 Before the Change

☐ Backup of the running PLC/SCADA taken
☐ Operations notified and agreed timing
☐ Bypasses or forces needed are recorded below
☐ MOC approved (if required)
☐ Equipment in a safe state for the work
☐ Test steps reviewed by a second person

Bypasses/forces applied (tag, reason, applied by): _____

K.5 Test Steps

Table K.1: Change Test Steps

#	Action	Expected result	Actual result	Pass / Fail	Init.
1	*Example:* With header pressure simulated at 40 torr, set `HMI_SP_Max_Vac_Press` to 45 torr and request the vacuum valve to open	Valve opens; permissive shown as met on SCADA			
2	*Example:* Raise simulated pressure to 50 torr and request the valve again	Valve does not open; permissive shown as not met with the reason			
3	*Example:* Enter 120 torr (above the hard-coded 100 torr maximum)	PLC clamps to 100 torr; SCADA shows the limit			
4					
5					
6					
7					

#	Action	Expected result	Actual result	Pass / Fail	Init.
8					

K.6 Regression and Abnormal-Condition Checks

Confirm that nothing connected to the change was affected. Check each item that applies:

☐ Normal operation of related equipment
☐ Alarms and first-out indication
☐ Mode changes (Off/Manual/Auto/Lockout)
☐ Operator security roles
☐ Bad-quality / out-of-range signal behavior
☐ Communication loss and recovery
☐ Power loss and restart behavior
☐ Other interlocks sharing the same inputs

K.7 Documentation Updated

☐ Electrical drawings ☐ I/O list ☐ Control Narrative
☐ Table of Interlocks and Permissives ☐ Alarm list ☐ Revision log / commit message
☐ Backup saved (after change) ☐ Released file name (no date) ☐ N/A

K.8 Return to Service and Sign-Off

☐ All bypasses and forces removed ☐ Equipment returned to normal mode ☐ Pre-startup safety review complete (if MOC)

Role	Name	Signature	Date
Performed by			
Independently verified by			
Accepted by Operations			